

ANTHONY SEBASTIAN SOSA LOROÑA

Full Stack Developer | Information Systems Engineering

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github.com/AnthonySosaL curricula-fawn.vercel.app atlas-lab-one.vercel.app

PROFESSIONAL SUMMARY

Information Systems Engineering graduate with Full Stack development, process automation and data analysis experience in government and private environments. Built ATLAS Lab, my own quantitative research lab with institutional-grade statistical validation (Deflated Sharpe, Monte Carlo, out-of-sample testing) over thousands of automatically evaluated strategies. I have built and deployed end-to-end systems: from data modeling and backend to the interface and its release on Linux servers. Specialized in turning manual processes into automated applications and in working on large legacy systems.

WORK EXPERIENCE

Nexus Soluciones S.A.S. B.I.C

Full Stack Developer (February 2026 – March 2026, Remote)

- Developed an e-signature SaaS platform with a two-application architecture, independent yet interconnected (client portal and distributor portal), each with its own backend and frontend.
- Built the backend with NestJS + Prisma ORM + PostgreSQL under a modular architecture with JWT authentication, covering the users, clients, plans, transfers, digital signatures, signed documents, discount codes and notifications modules.
- Developed the frontend with Next.js 16 + React 19 + TypeScript + Tailwind CSS, including dashboards, dynamic tables and metrics visualization.
- Integrated AWS S3 for storing documents and digital certificates, and the Payphone payment gateway with a discount code system.
- Deployed both applications on Railway with Docker, ensuring consistent environments.

Ministry of Transport and Public Works (Ecuador)

Full Stack Developer · Infrastructure Team (December 2024 – June 2025, Hybrid)

- Automated the national technology infrastructure inventory: turned a survey scattered across Excel sheets into an internal web application (Java backend, JavaScript frontend) to register and query VoIP equipment, desktops, laptops, racks, routers, switches and connection points, eliminating manual file handling.
- Coordinated, with a team of 4, the data cleanup, moving from manual cleaning to a reproducible process with Python and Jupyter Notebook.
- Used the Cisco management console to locate and verify specifications of devices whose records did not match or that were in areas without physical access, improving the accuracy of the final inventory.
- Started the role providing technical support and doing field surveys, which gave me the operational context to design the solution that later automated the process.
- Configured and deployed the application on an institutional Linux server.

Ministry of Telecommunications and Information Society (Ecuador)

Full Stack Developer · Development Area (July 2024 – September 2024, On-site)

- Developed and deployed an internal training web platform for Ministry staff, fully configuring the server where it was hosted. Added usage metrics (logins per user and interaction per course) that made it possible to identify which content was actually being used.
- Automated the issuance of certifications (Cisco, Claro, Movistar) and file consolidation using Excel and Word macros, eliminating the manual generation of each certificate.
- Fixed bugs and optimized SIADI, a legacy system from an external vendor made up of 3 backends and 2 frontends (React, TypeScript, PostgreSQL), working on code not developed in-house.
- Developed the SIADI 2.0 prototype in Java, with separate backend and frontend modules, adapting the features to the Ministry's specific requirements.
- Managed Docker containers and deployed on AlmaLinux with SSL certificates.

Helios Trader Group LLC

Financial Markets Specialist (2021 – 2022, Remote)

- Passed the selection process and managed an investment account applying my own risk management strategy.
- Prepared monthly performance reports and adjusted the strategy based on risk and return metrics.

PROJECTS

ATLAS Lab · Quantitative research platform · atlas-lab-one.vercel.app

- Designed and built end to end an algorithmic trading research lab: data pipeline (ingestion from market APIs, cleaning and storage of 27 instruments with 12+ years of history), Python analysis engine (pandas, numpy) and a Next.js + TypeScript frontend with interactive visualizations.
- Implemented institutional-grade statistical validation (Deflated Sharpe, Monte Carlo simulations, out-of-sample testing) on 34,751 automatically generated and evaluated strategies.
- Automated daily monitoring through scheduled tasks with failure recovery, and integrated the system with MetaTrader 5 for execution in a test environment.
- Used data sources such as Dukascopy, yfinance, FRED and CFTC, with Parquet storage (PyArrow) and statistical analysis with scipy and scikit-learn, plus my own MCP server to operate the system.
- Built an experiments dashboard in Flask to review the lab's results and metrics.
- The site publishes all results, including negative ones, as an exercise in methodological transparency.

Certis · English assessment and practice platform (CEFR) · english-c1.runasp.net

- Designed and built a complete English assessment and practice platform under the CEFR framework (A2-C1), with an ASP.NET Core 8 (C#) backend using a layered architecture (Domain/Application/Infrastructure/Api) and Entity Framework Core on SQL Server, authentication with ASP.NET Core Identity and JWT, and an Angular 22 + Angular Material frontend.
- Implemented a level-based assessment algorithm with a question bank (Grammar, Vocabulary, Reading, Listening) and AI-powered Speaking practice (Groq) in Cambridge exam-style scenarios.
- Integrated AI-generated questions and personalized feedback (Groq, structured JSON output), and local audio generation for Listening with edge-tts.
- Built an admin/tutoring panel with content management, role-based access control, and early alerts on student performance.
- Deployed the backend and frontend to production (MonsterASP.NET) with a real SQL Server database, with domain unit tests and end-to-end verification on every change.

JARVIS · Desktop AI assistant for Windows · Node.js, PowerShell/Win32, multi-provider LLMs

- Designed and built an assistant that controls Windows by voice and text: LLM intent classification pipeline, multi-provider router (Groq, Gemini, NVIDIA NIM) with automatic failover, a self-correcting agentic loop, and a plugin architecture with hot-reload that generates new skills at runtime.
- ~9,000 lines of original code (Node.js + Express backend, operating system control via PowerShell and Win32 API), with a suite of 36 integration tests at 100% pass rate.

Real-time meeting assistant · Python, local Whisper on GPU, multi-LLM router

- Developed a desktop assistant that transcribes live audio with GPU-accelerated local Whisper (~0.3 s latency) and queries multiple AI providers (Groq, Gemini, NVIDIA) through my own router with cascading fallback (GPU → CPU → cloud), so it never runs out of answers.

YouTube channel automation and analytics system · Python, Telegram Bot, YouTube Data/Analytics API

- Built a content production pipeline controlled by a Telegram bot (start and stop, mode switching, hot-editable quotas, live logs), with data extraction via YouTube Data API v3 and Analytics API v2 over a history of 16,000+ videos, and dashboards in Flask and Obsidian.
- Used that data to make real product decisions: detected that the views distribution was abnormally flat, ruled out retention as a bottleneck, measured the real upload ceiling per channel, and built a topic prioritization model with statistical validation (train/test split).

Curricula · Full Stack portfolio with AI · curricula-fawn.vercel.app

- Developed a complete platform with React 19, NestJS and PostgreSQL (Neon), with a Vite, TypeScript and Tailwind frontend and 3D visualization (Three.js), deployed to production (Vercel + Render).
- Integrated an AI chat (Groq) and an analytics panel with automatically generated recommendations, plus a mini-game with its own leaderboard.
- Implemented JWT authentication, rate limiting and security headers, actively verified against the production server.

CERTIFICATIONS

- AWS Academy Graduate – Cloud Foundations (Amazon Web Services).
- AWS SimuLearn – Cloud Practitioner (Amazon Web Services).
- AWS SimuLearn – AI Practitioner (Amazon Web Services).

EDUCATION

- Information Systems Engineering degree, Pontificia Universidad Católica del Ecuador (PUCE) / 2022 – 2026. Graduated.
- High School Diploma (Bachillerato General Unificado), Unidad Educativa Santo Domingo de Guzmán / 2008 – 2021.

SKILLS

- Languages: Java, JavaScript, TypeScript, Python, C#, SQL, HTML5, CSS3.
- Frameworks/Libraries: React, Next.js, NestJS, Spring Boot, Node.js, FastAPI, Flask, .NET / ASP.NET Core (personal project in production: Certis), Angular (personal project in production: Certis), React Native, Flutter, Tailwind CSS, shadcn/ui, Radix UI.
- Interface design: Figma, Adobe XD, Sketch, Adobe Photoshop.
- Audiovisual production: Adobe Premiere, After Effects, Blender.
- Databases: PostgreSQL, Oracle (professional production); SQL Server (personal project in production: Certis); MySQL, Firebase, MongoDB · Prisma ORM.
- Big Data / ETL (academic training, final semesters, business cases): Hadoop, Apache Spark, HDFS, MapReduce.
- Cloud / DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB, VPC, IAM, Elastic Load Balancing, Auto Scaling, Route 53, CloudFront, CloudWatch, CloudTrail, CloudFormation, AWS Well-Architected Framework; AWS Academy and AWS SimuLearn training); Docker, Linux (AlmaLinux), SSL, Railway, Vercel.
- Data and automation: Python (pandas, numpy, scipy), PyArrow/Parquet, Jupyter Notebook, n8n (workflow automation, personal projects), advanced Excel, Excel and Word macros, web scraping (BeautifulSoup, Selenium).
- Machine Learning / Deep Learning (academic training, Data Science courses): scikit-learn, TensorFlow, PyTorch.
- Infrastructure and networking: Linux server administration, Bash and PowerShell scripting (cron jobs, server and OS automation), Cisco console, infrastructure inventory (racks, switches, routers, VoIP).
- AI applied to development: Claude Code (daily use), Codex (in use since the Ministry of Telecommunications, 2024), Gemini, Perplexity, OpenAI API, Groq API, NVIDIA API (free models for automations), local models (Llama, Mistral) via LM Studio, MCP (Model Context Protocol), Whisper, NLP, LLM pipelines, multi-provider routers with failover.
- Generative AI on AWS (AWS SimuLearn AI Practitioner training): Amazon Bedrock, Amazon Bedrock Runtime, Amazon Nova Pro, Amazon SageMaker Studio, Amazon Q Developer, RAG (Retrieval-Augmented Generation), Bedrock Knowledge Bases, vector databases, embeddings, tool calling, prompt engineering, Amazon Textract, Amazon Comprehend, Amazon Translate, Amazon Polly, Amazon Transcribe, Boto3.
- Version control and methodologies: Git, GitHub, Kanban, Scrum.
- Soft skills: problem solving, teamwork, communication with technical and business teams, adaptability.

LANGUAGES

- Spanish: Native.
- English: Intermediate (B1).